

Bethesda System for Reporting Thyroid Cytopathology

Standardized cytologic diagnosis for thyroid FNA

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BAMEDICAL
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Introduction

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- Fine-needle aspiration (FNA) is among the first diagnostic tools used in the evaluation of thyroid nodules
- Thyroid FNA offers high positive predictive value (97%–99%), with sensitivities 65% to 99% and specificities of 72% to 100%
- Many potential diagnostic pitfalls exist that can lead to false-positive (0-28%) and/or false-negative (1-35%)

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HOW TO MAKE A GOOD DIAGNOSIS ?

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Bethesda System for Reporting Thyroid Cytopathology

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BSRTC is the internationally accepted classification to communicate cytological diagnosis in a uniform and unambiguous way

Table 1.1 The Bethesda System for Reporting Thyroid Cytopathology; diagnostic categories

I. Nondiagnostic
Cyst fluid only
Virtually acellular specimen
Other (obscuring blood, clotting artifact, drying artifact, etc.)
II. Benign
Consistent with follicular nodular disease (includes adenomatoid nodule, colloid nodule, etc.)
Consistent with chronic lymphocytic (Hashimoto) thyroiditis in the proper clinical context
Consistent with granulomatous (subacute) thyroiditis
Other
III. Atypia of Undetermined Significance
Specify if AUS-nuclear atypia or AUS-other
IV. Follicular Neoplasm
Specify if oncocytic (Hürthle cell) type
V. Suspicious for Malignancy
Suspicious for papillary thyroid carcinoma
Suspicious for medullary thyroid carcinoma
Suspicious for metastatic carcinoma
Suspicious for lymphoma
Other
VI. Malignant
Papillary thyroid carcinoma
High-grade follicular cell-derived non-anaplastic thyroid carcinoma
Medullary thyroid carcinoma
Undifferentiated (anaplastic) carcinoma
Squamous cell carcinoma
Carcinoma with mixed features (specify)
Metastatic malignancy
Non-Hodgkin lymphoma
Other

Risk of Malignancy and Management

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Diagnostic category	ROM ^a	Usual management ^b
	Mean% (range)	
Nondiagnostic	13 (5–20) ^c	Repeat FNA ^d with ultrasound guidance
Benign	4 (2–7) ^e	Clinical and sonographic follow-up
Atypia of Undetermined Significance ^f	22 (13–30)	Repeat FNA ^d , molecular testing, diagnostic lobectomy, or surveillance
Follicular Neoplasm ^g	30 (23–34)	Molecular testing ^h , diagnostic lobectomy
Suspicious for Malignancy	74 (67–83)	Molecular testing ^h , lobectomy or near-total thyroidectomy ⁱ
Malignant	97 (97–100)	Lobectomy or near-total thyroidectomy ⁱ

Criteria for Adequacy

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- A thyroid FNA sample is considered adequate for evaluation if it contains a minimum of **six groups of well-visualized (i.e., well stained, well preserved, undistorted, and unobstructed) follicular epithelial cells with at least ten cells per group.**
- These six groups of ten follicular cells could be either on one slide or distributed among several for adequacy determination.
- These criteria apply to all cytologic preparations (including conventional smears and liquid-based samples).

Exceptions to Adequacy Criteria

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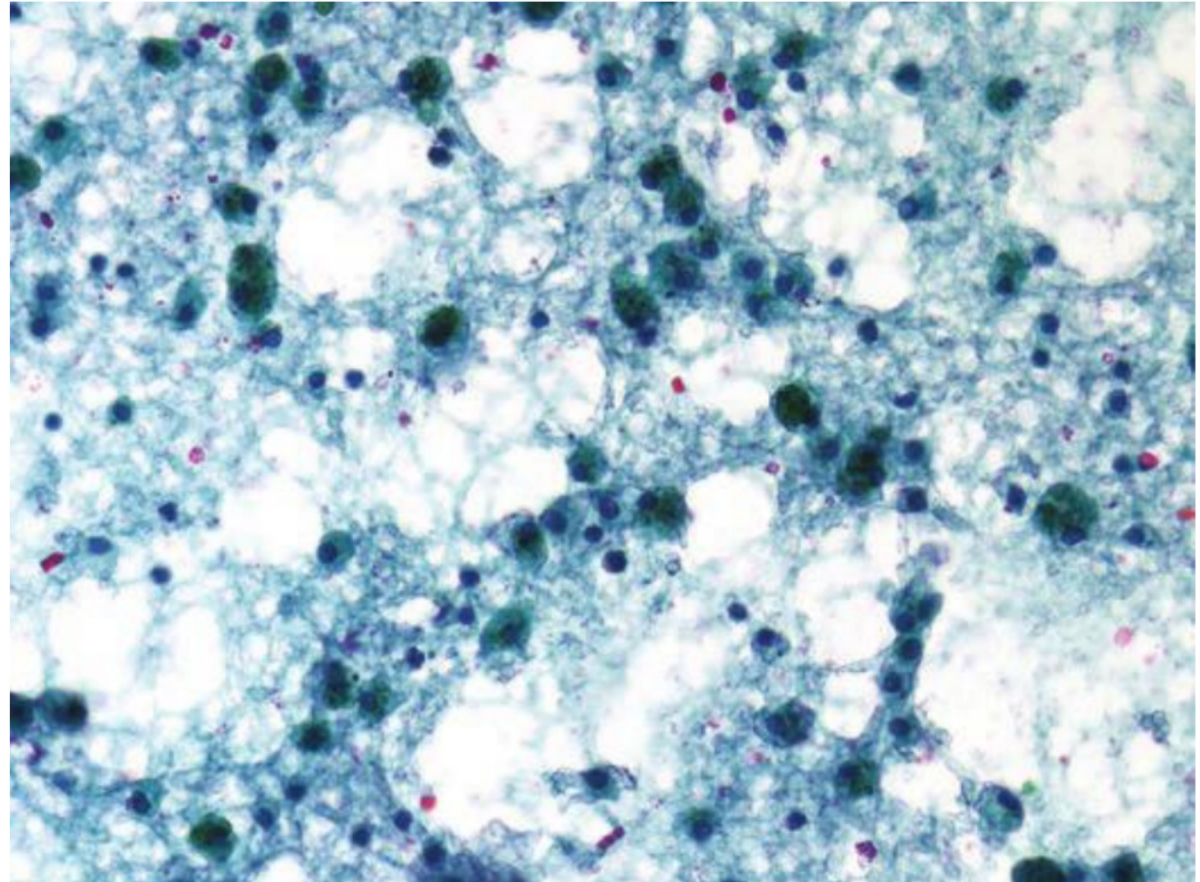
Exceptions to this requirement apply to a limited number of case types, including:

- ❖ Aspirates with cytologic atypia
- ❖ Solid nodules with inflammation
- ❖ Colloid nodules

1. Non-diagnostic / Unsatisfactory

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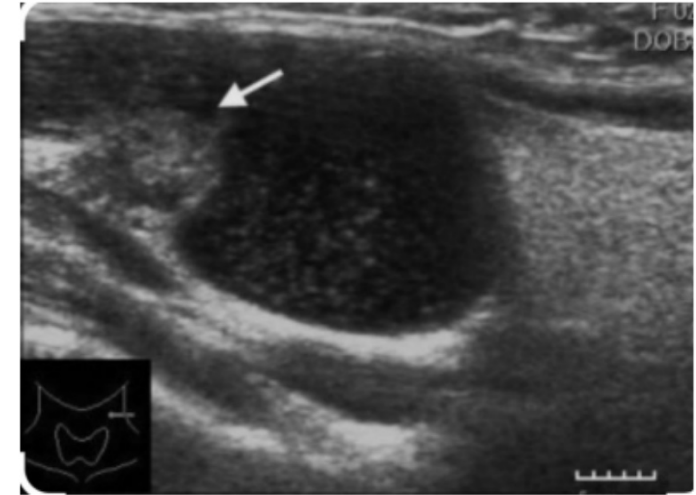
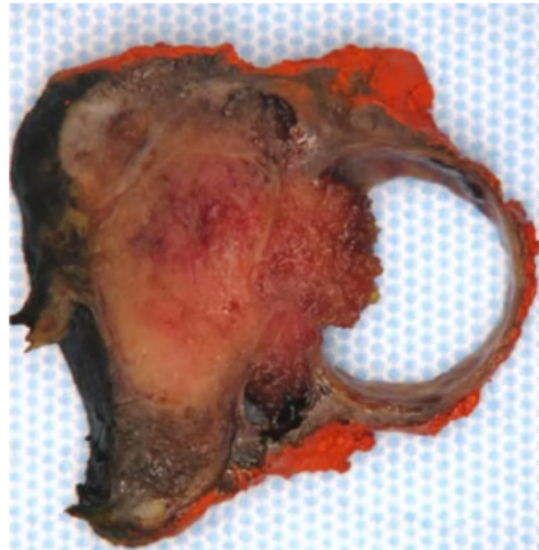
- Incidence : 10%
- Do not meet the criteria for adequacy
- Recommendation : FNA with USG guidance



Pitfalls: Cystic Papillary Thyroid Carcinoma

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- Rule out papillary thyroid carcinoma, cystic variant
- Up to 10–16 % of PTC can be cystic
- In cystic change, malignant tumor cells commonly **misinterpreted as macrophages (non diagnostic)**
→ false negative



Importance of USG-guided FNA

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The importance of USG guided FNA

- Retrospective studies : **FNA using USG guidance → lower rates of nondiagnostic and false-negative** compared to palpation

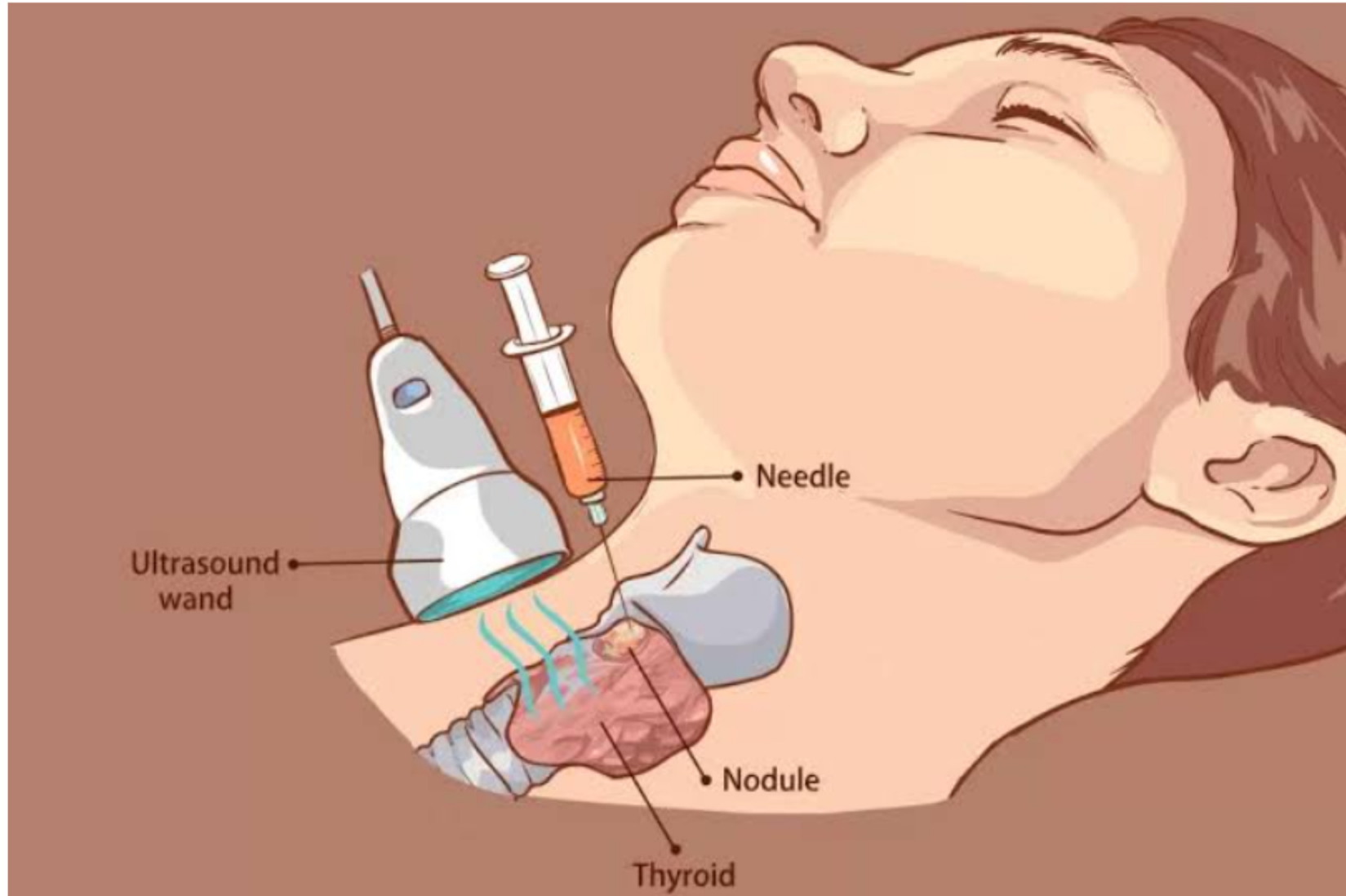
Table 3: Inadequate biopsy rates in studies that compared palpation-guided versus ultrasound-guided thyroid fine-needle aspiration biopsies

Author and year	No. of nodules	Palpation-guided	Ultrasound-guided	Reference
Danese 1998	9683	9%	4%	[11]
Hatada 1998	166	30%	17%	[12]
Carmeci 1998	497	16%	7%	[5]
Mehrotra 2005	262	47%	16%	[8]
Ghofrani 2006	1502	17%	7%	[9]
Cai 2006	434	13%	6%	[13]
Cesur 2006	285	32%	21%	[7]
Izquierdo 2006	376	11%	7%	[10]
Can 2008	386	27%	13%	This study

BMC Research Notes 2008, 1:12 <http://www.biomedcentral.com/1756-0500/1/12>

USG-guided FNA Technique

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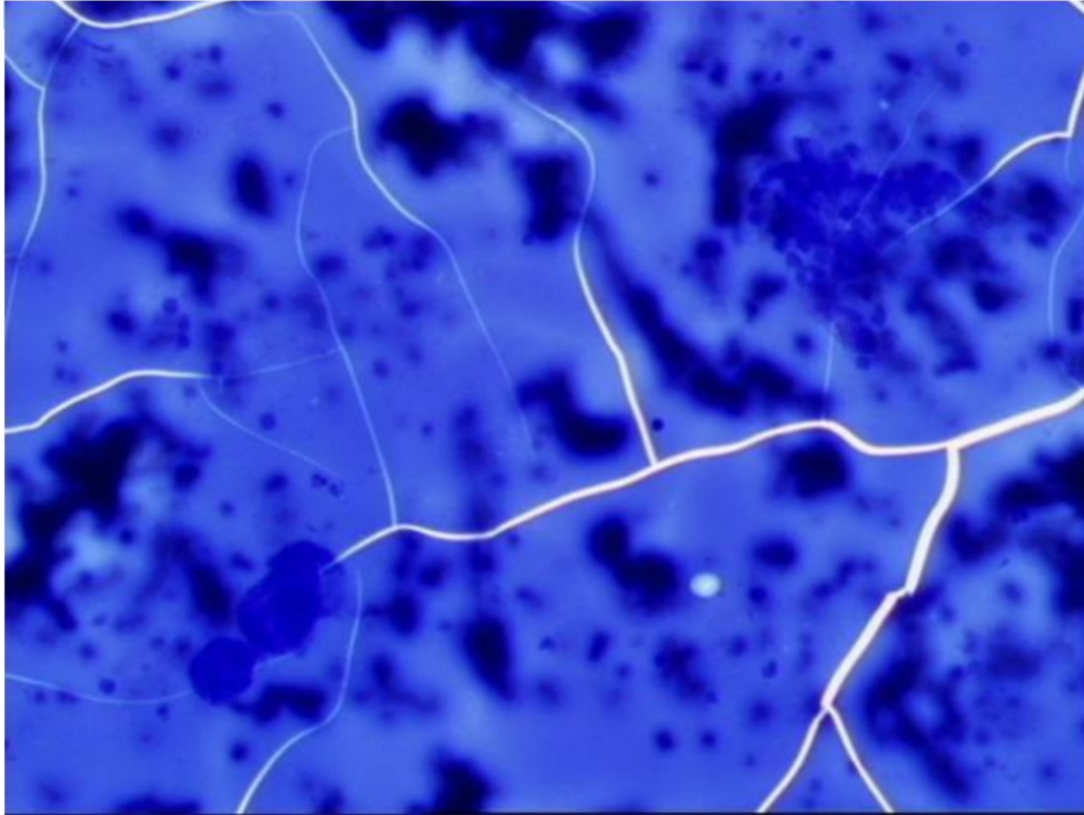
2. Benign

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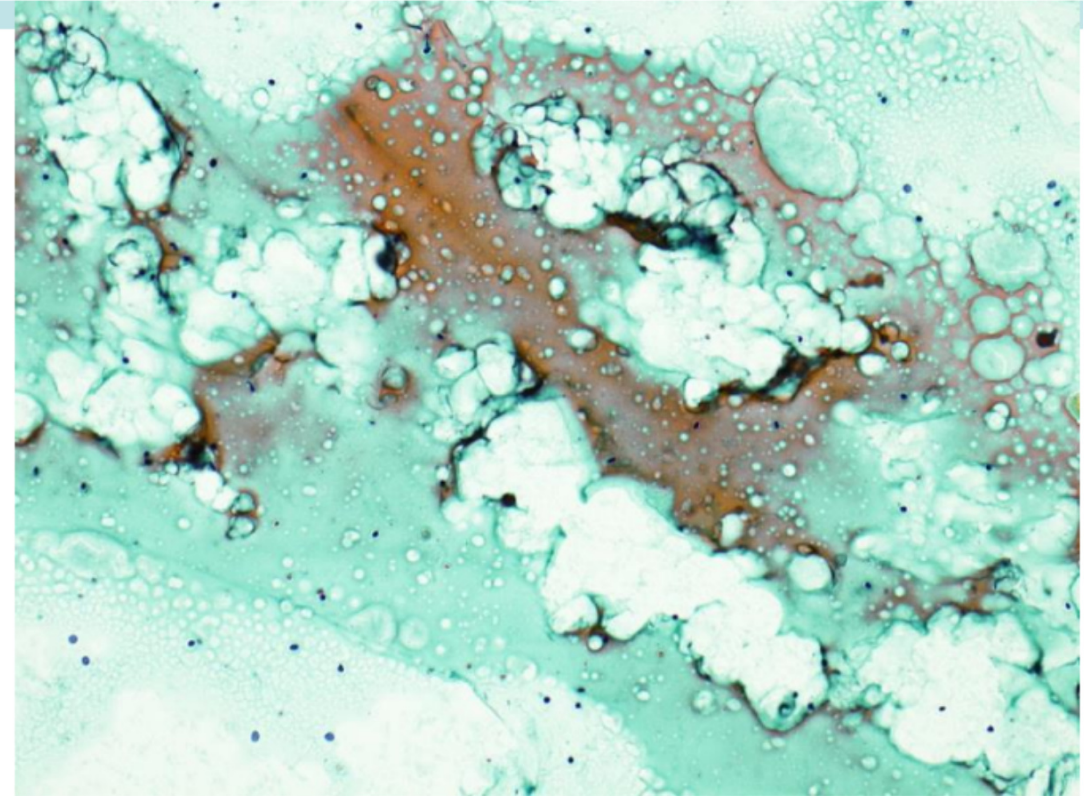
- ❖ Adequate for evaluation and consists of colloid and benign-appearing follicular cells in varying proportions
- ❖ Included entities:
 - ❖ Benign follicular nodule:
 - ❖ Colloid nodules
 - ❖ Nodular goiter
 - ❖ Hyperplastic (adenomatoid) nodules
 - ❖ Nodules in Graves' disease
 - ❖ Thyroiditis
 - ❖ Lymphocytic thyroiditis
 - ❖ Granulomatous thyroiditis

Benign Cytology: Colloid & Follicular Cells

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Thick colloid



watery colloid

Benign Cytomorphologic Features

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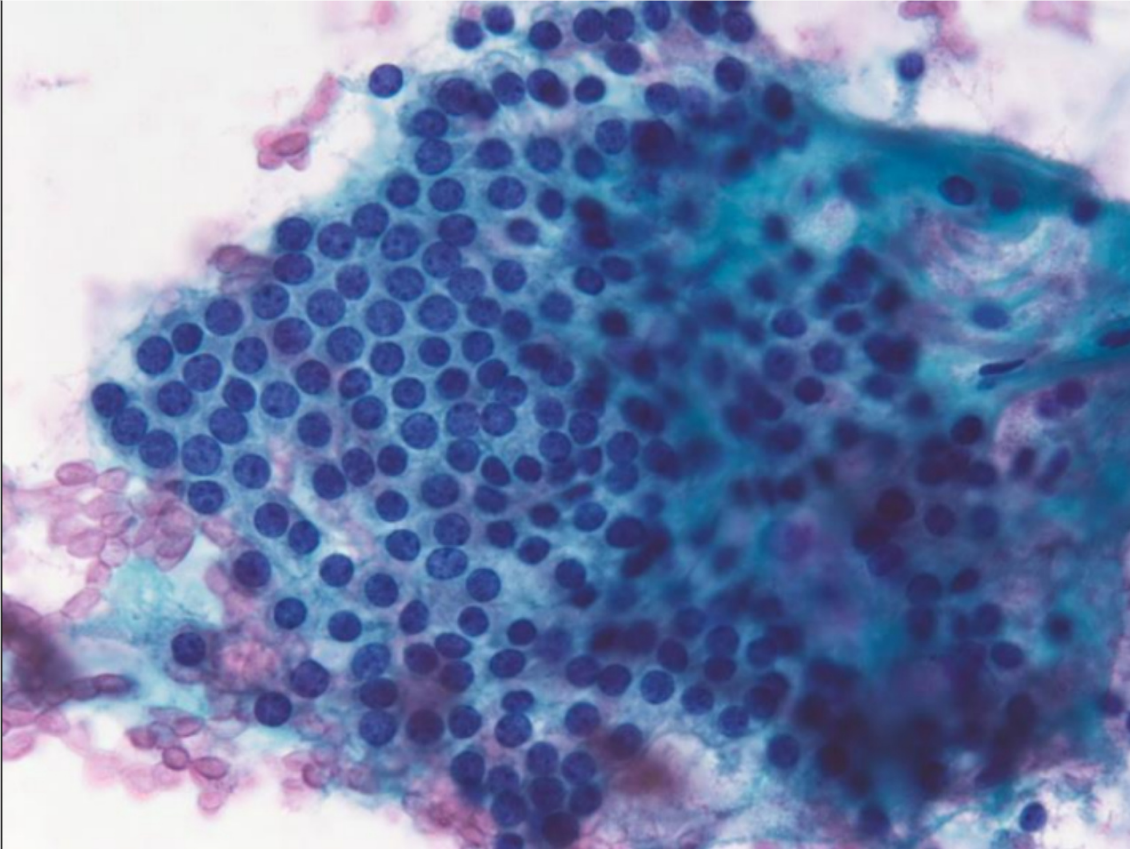
- Monolayered sheets of follicular cells
- Occasional three dimensional balls/spheres
- Evenly spaced nuclear arrangement
- Follicular cell nuclei are round to oval, approximately the size of a red blood cell
- Colloid background

Benign Cytology: Microscopy

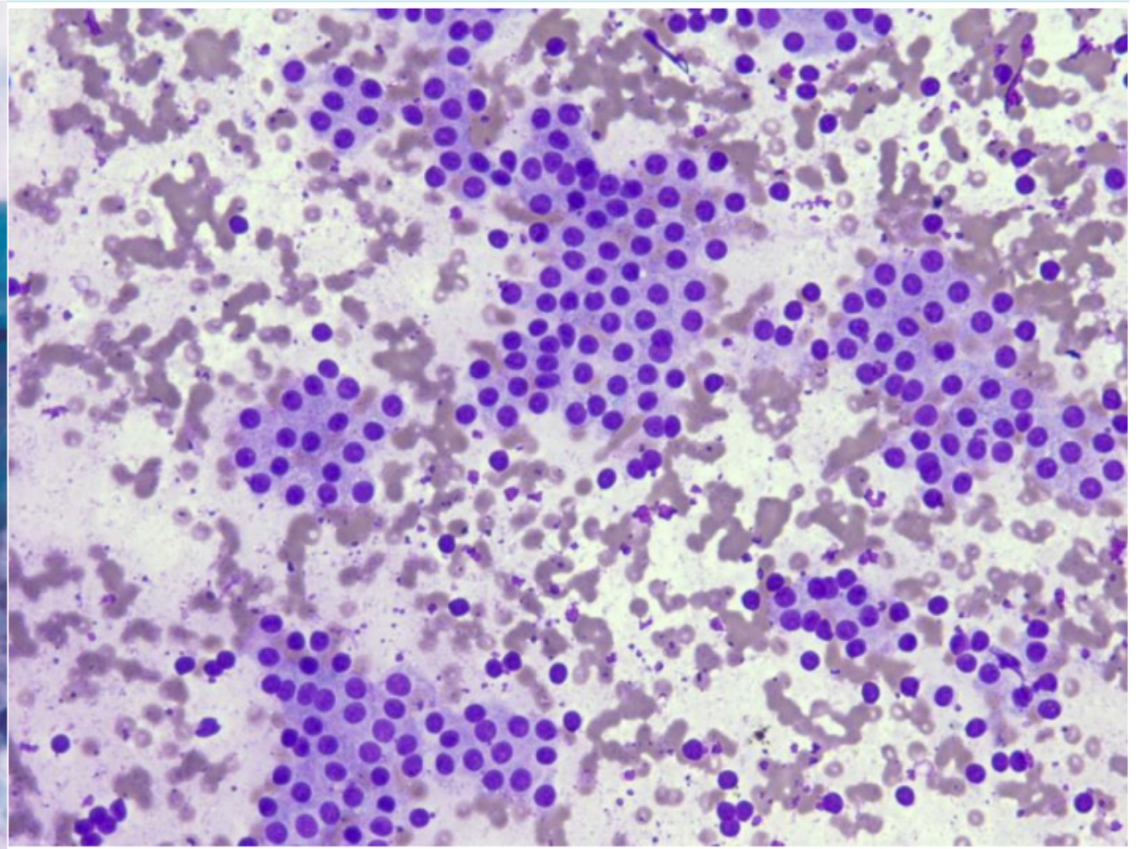
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Minimal nuclear overlapping, no nuclear pallor nor atypia

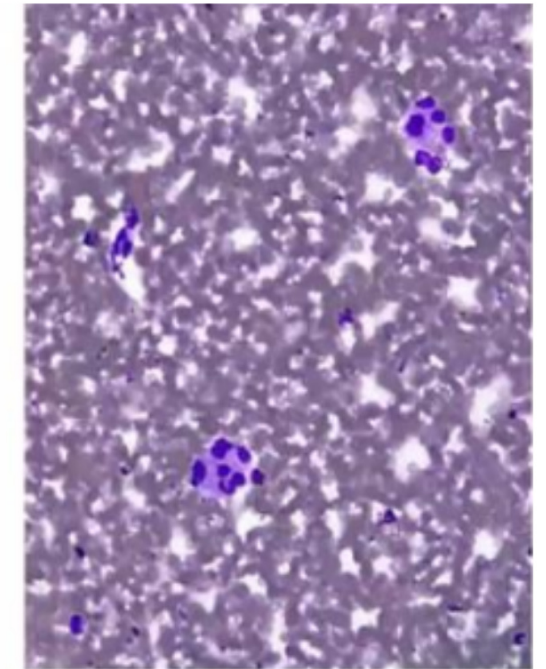
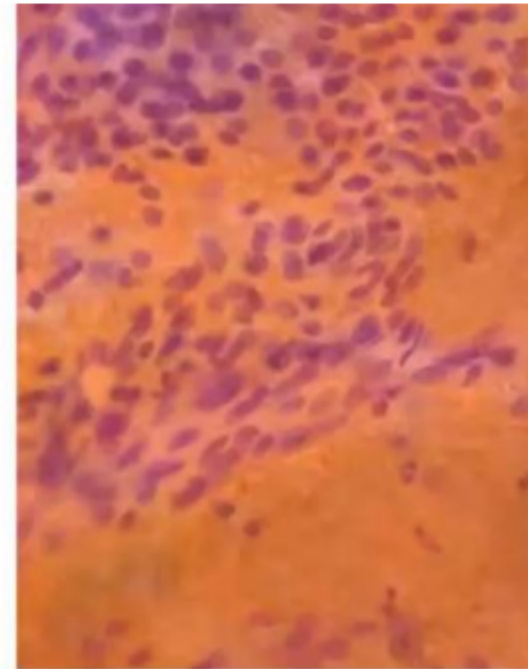
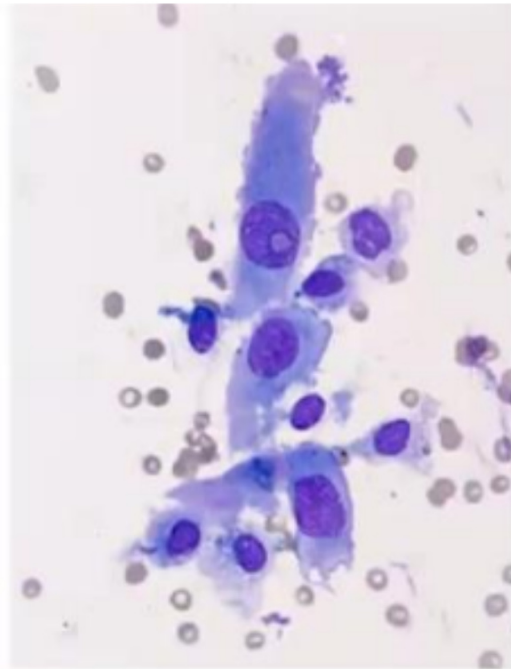
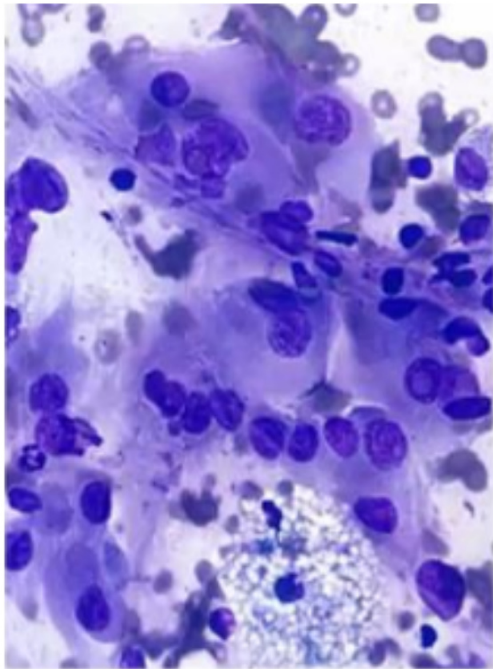


Microfollicles may be present but not prominent

3. Atypia of Undetermined Significance (AUS)

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Specimens that contain cells (follicular, lymphoid, or other) with architectural and/or nuclear atypia that is not sufficient to be classified as suspicious for a follicular neoplasm, suspicious for malignancy, or malignant; but more marked than can be ascribed confidently to benign changes.



4. Follicular Neoplasm

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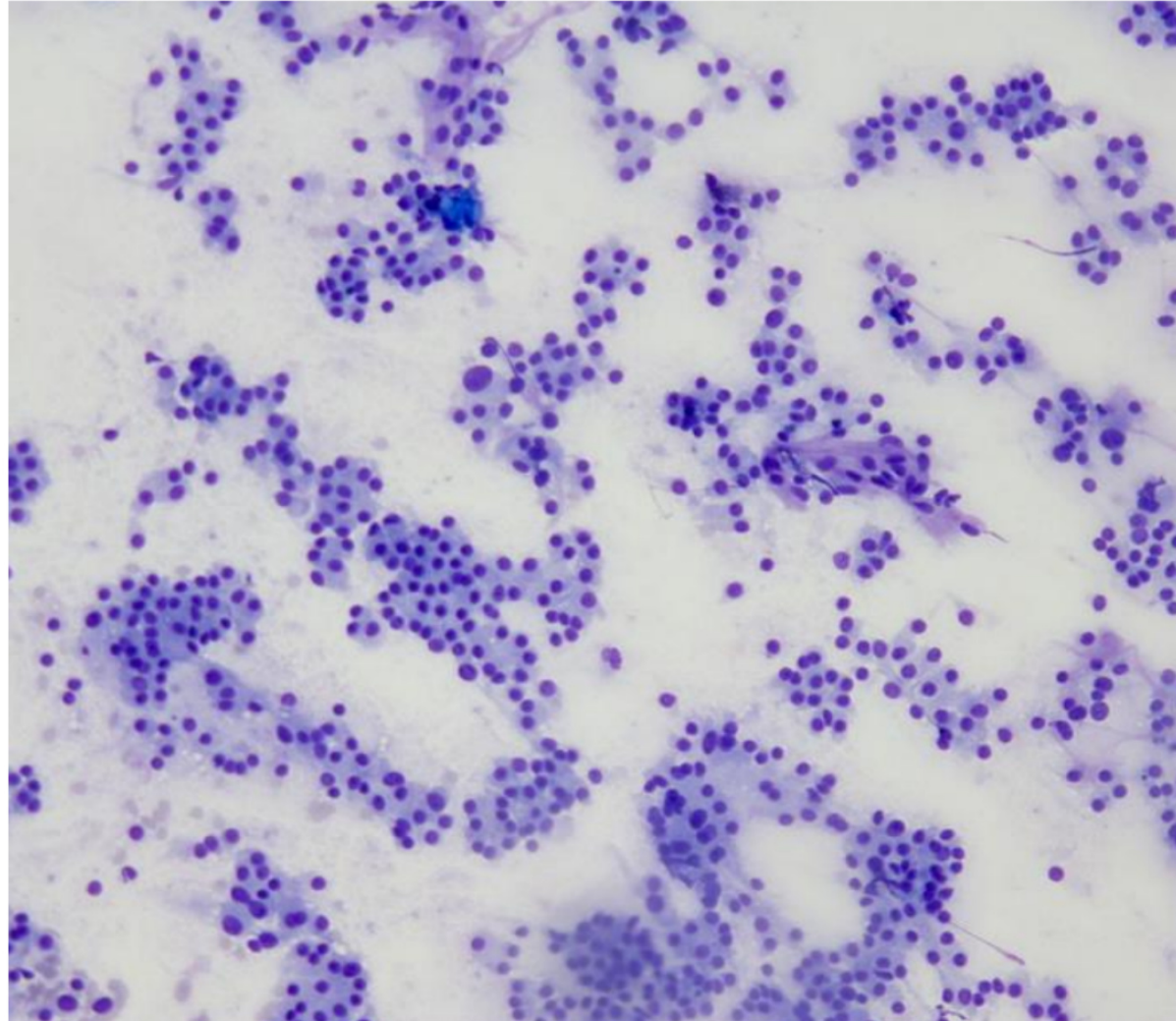
- ROM 25-40%
- Significant architectural atypia
- Predominant microfollicles and trabeculae
- Cellular smear
- Mild nuclear change may be seen (no INCIIs or papillae)
- Colloid scant or absent
- Lobectomy is necessary for definitive diagnosis

Follicular Neoplasm: Cytology

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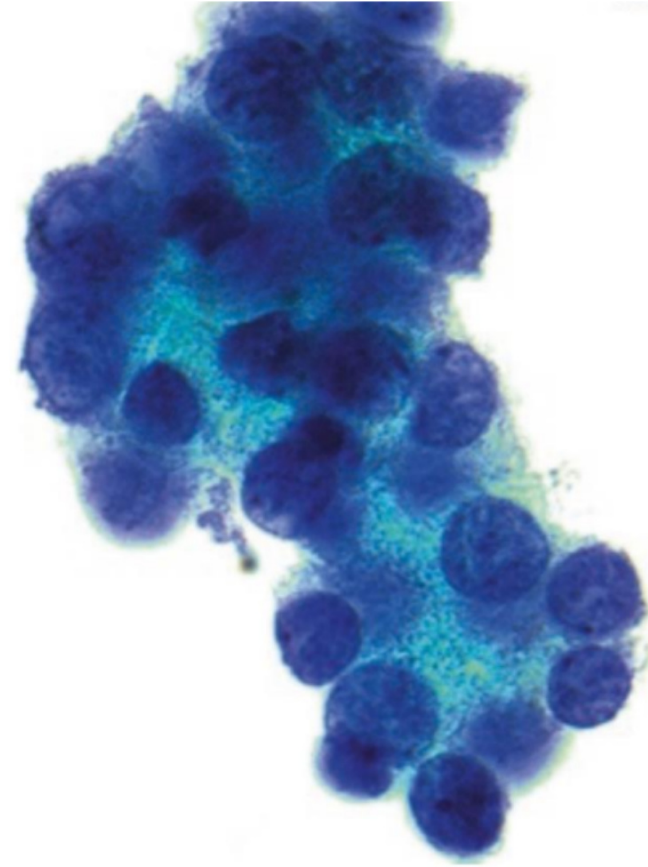
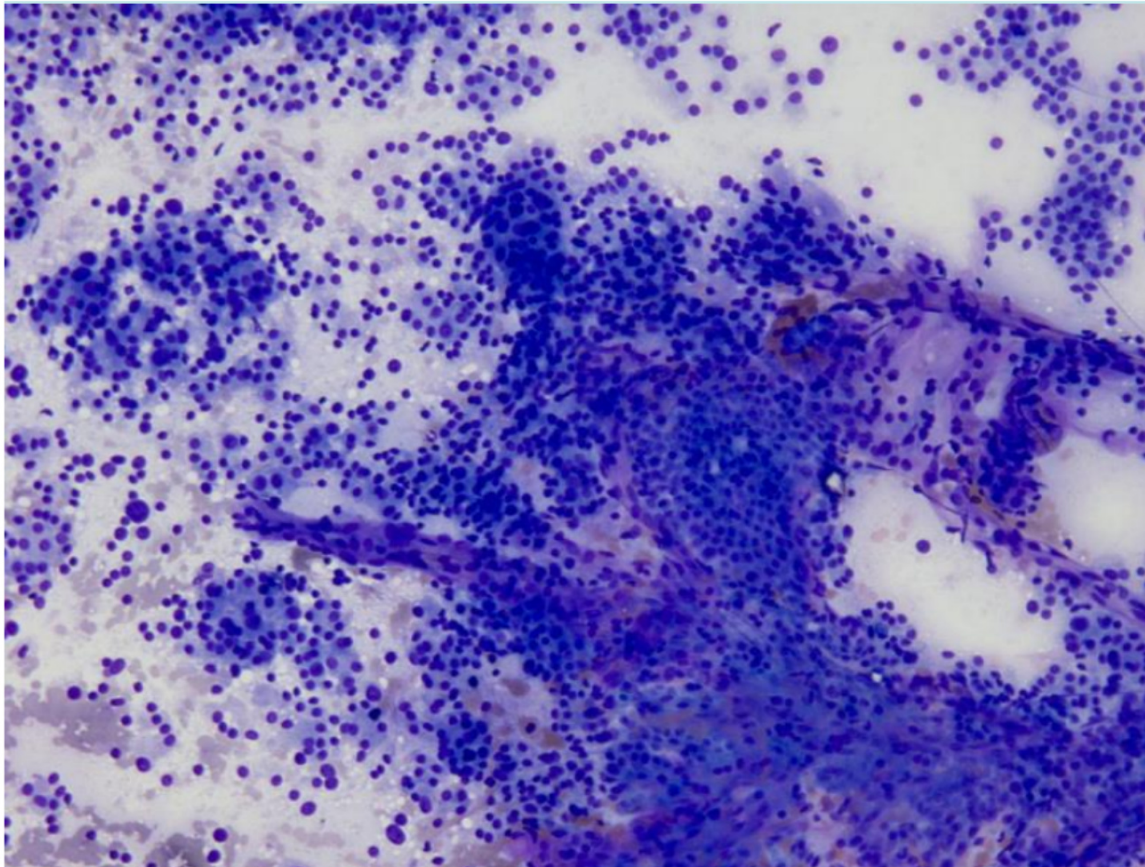


Follicular Neoplasm: Cellular Smears

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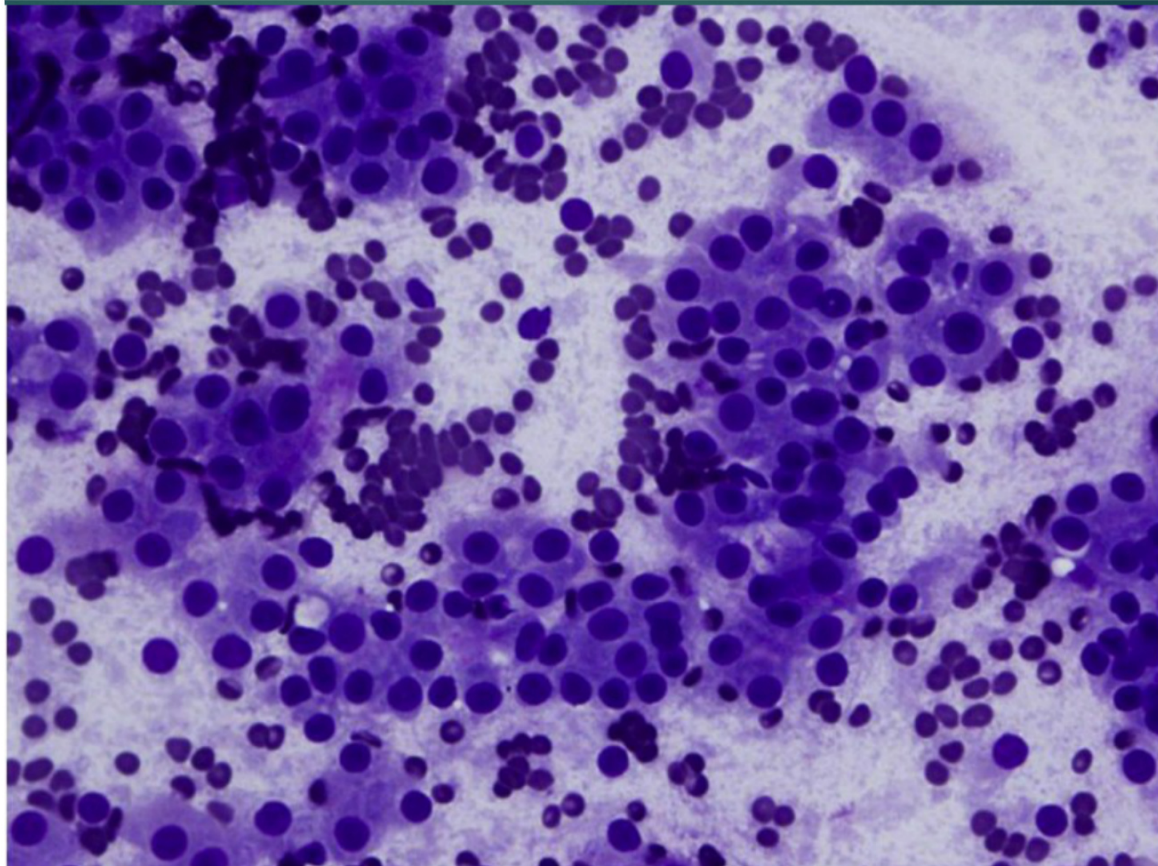
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Follicular Neoplasm, Hürthle Cell Type

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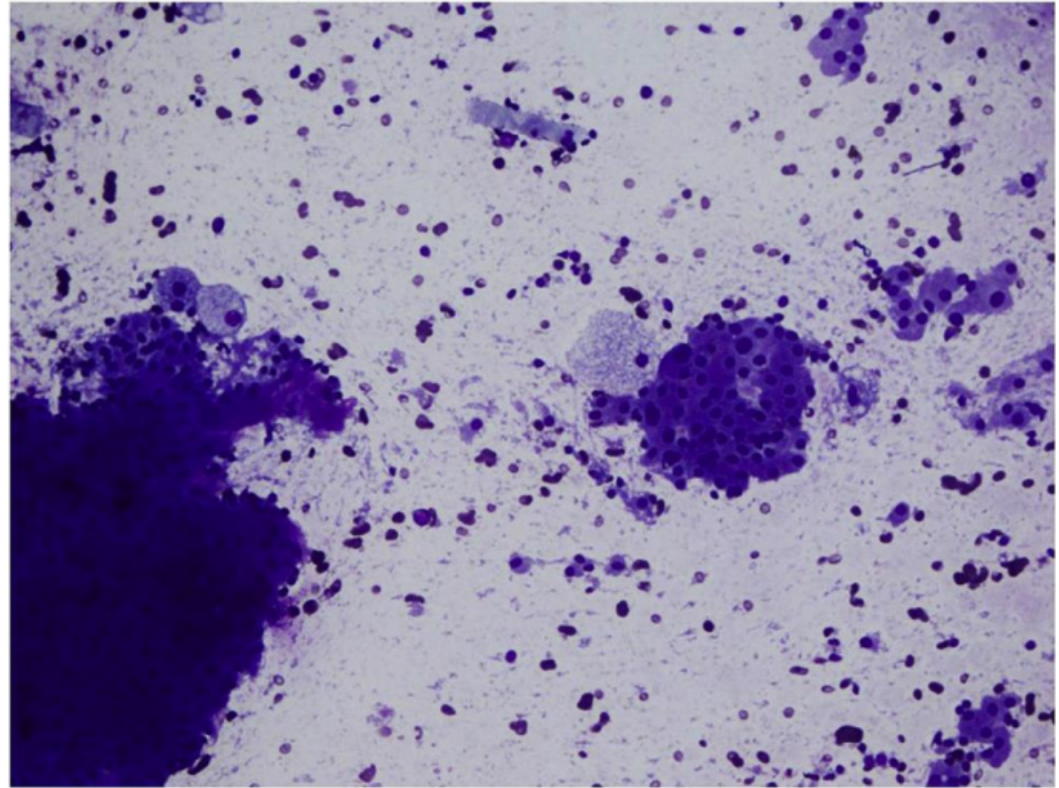
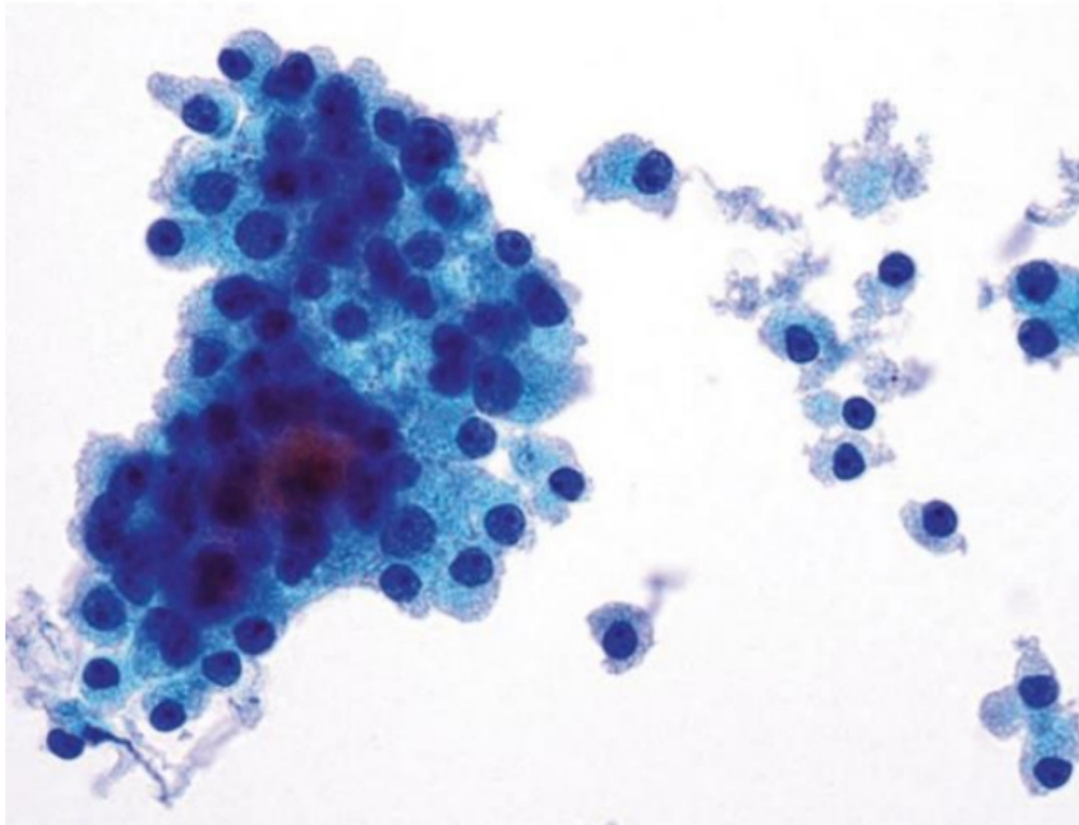
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Hürthle Cell Type: Cytology

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5. Suspicious for Malignancy

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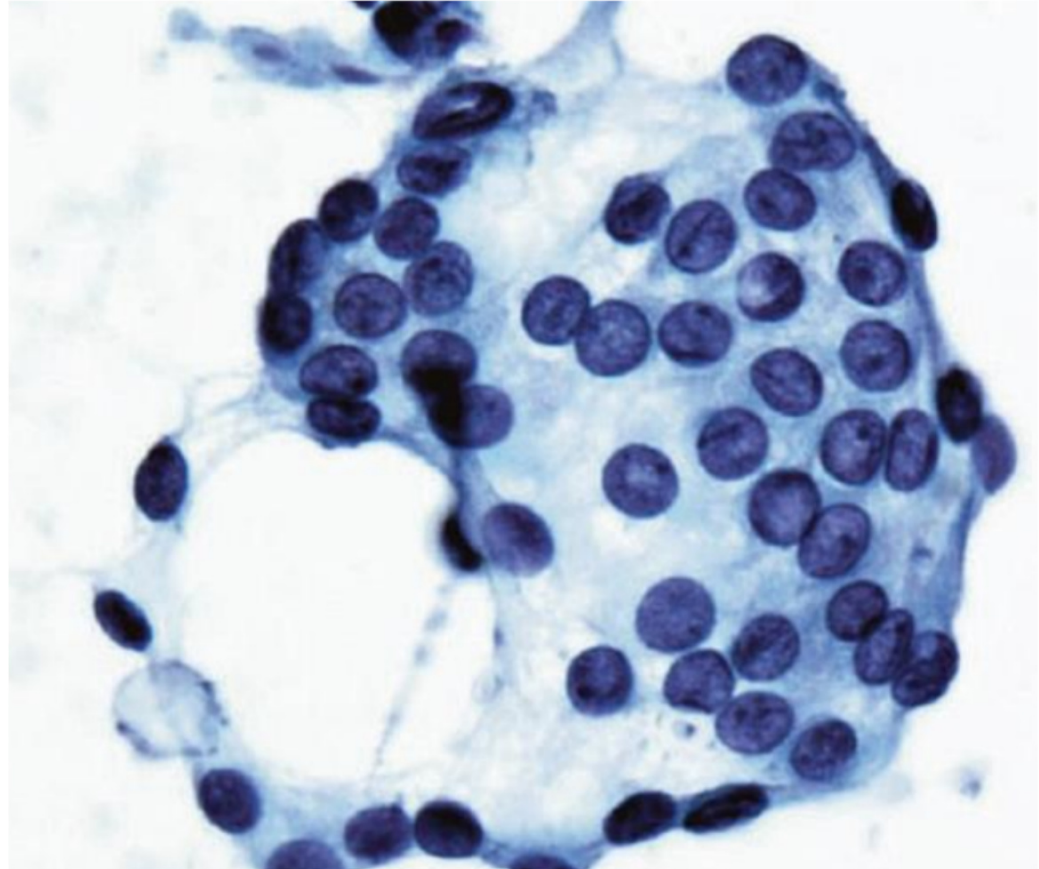
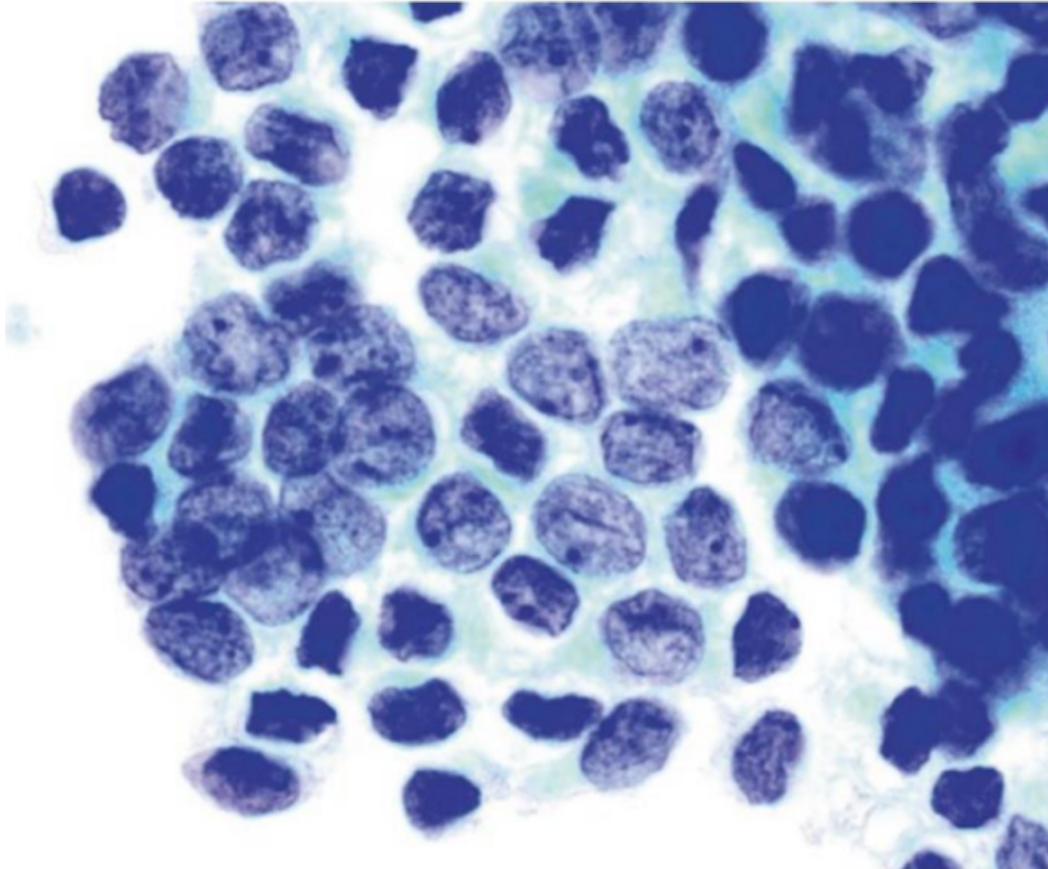
- ❖ ROM: 65-75%
- ❖ Cytomorphologic features (most often those of PTC) raise a strong suspicion of malignancy but the findings are not sufficient for a conclusive diagnosis
- ❖ Included entities:
 - ❖ Suspicious for papillary carcinoma (FVPTC/NIFTP may include in this category)
 - ❖ Suspicious for medullary carcinoma
 - ❖ Suspicious for lymphoma
 - ❖ Suspicious for metastatic

Suspicious for Malignancy: Cytology

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6. Malignant

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Includes :

- Papillary carcinoma
- Medullary carcinoma
- Lymphoma
- Metastatic tumors

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Malignant Cytology: Papillary Carcinoma

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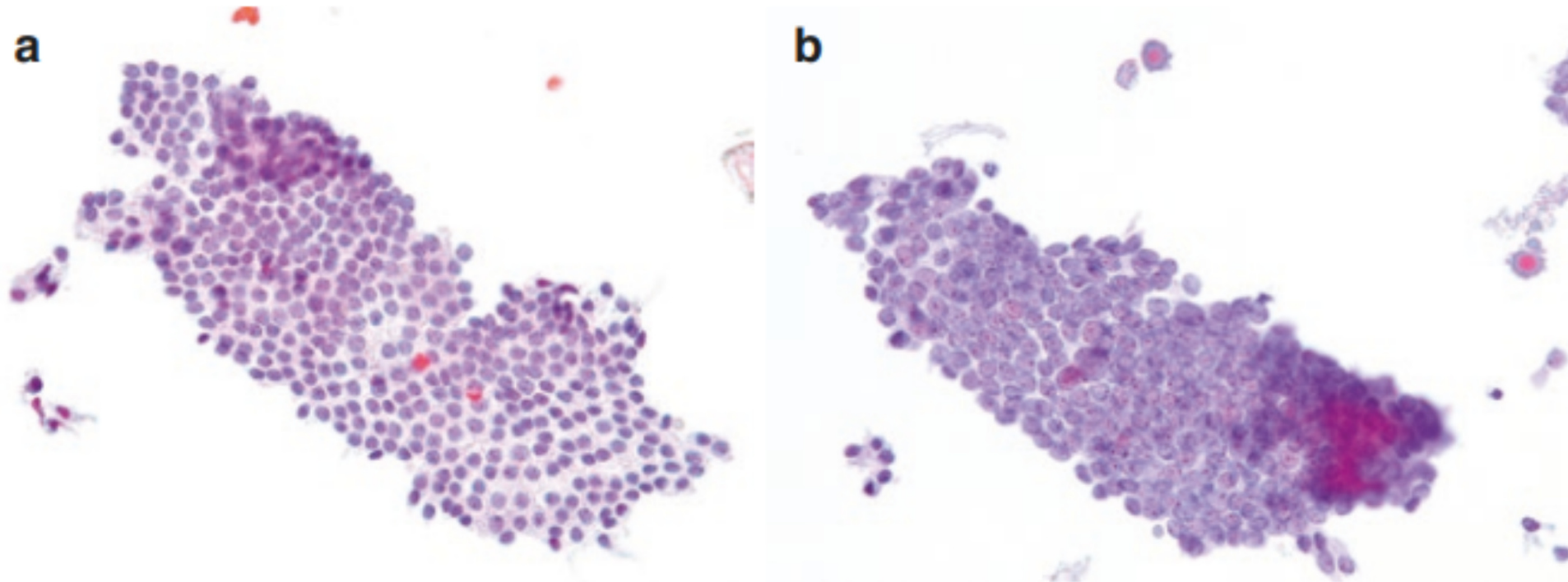


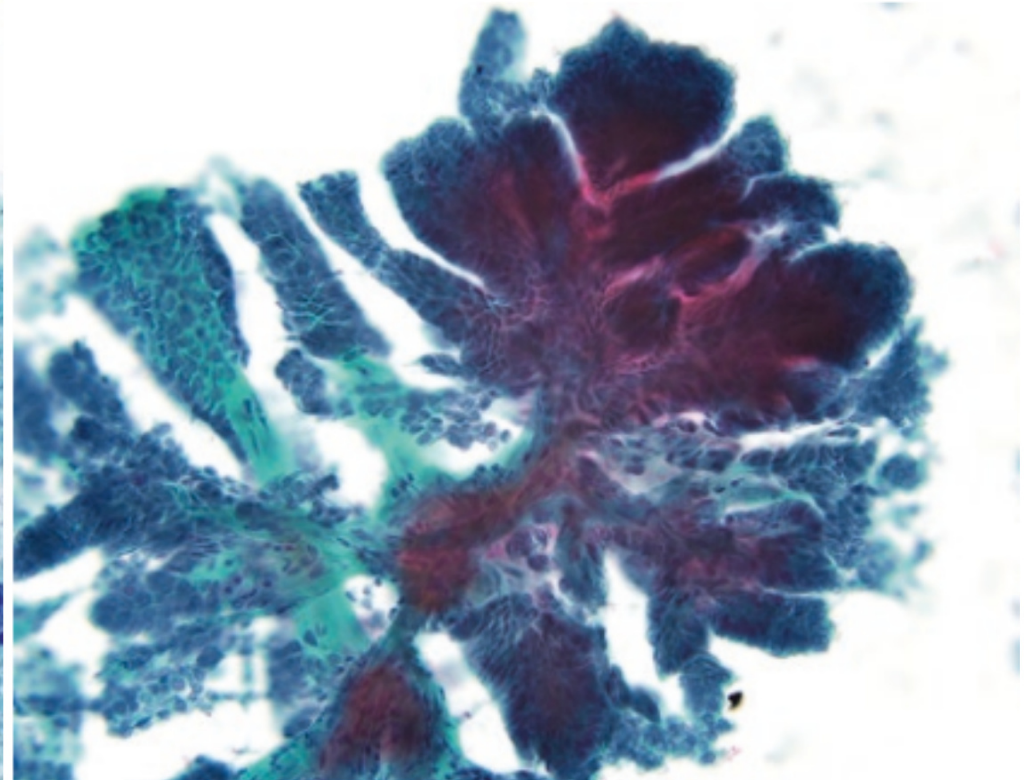
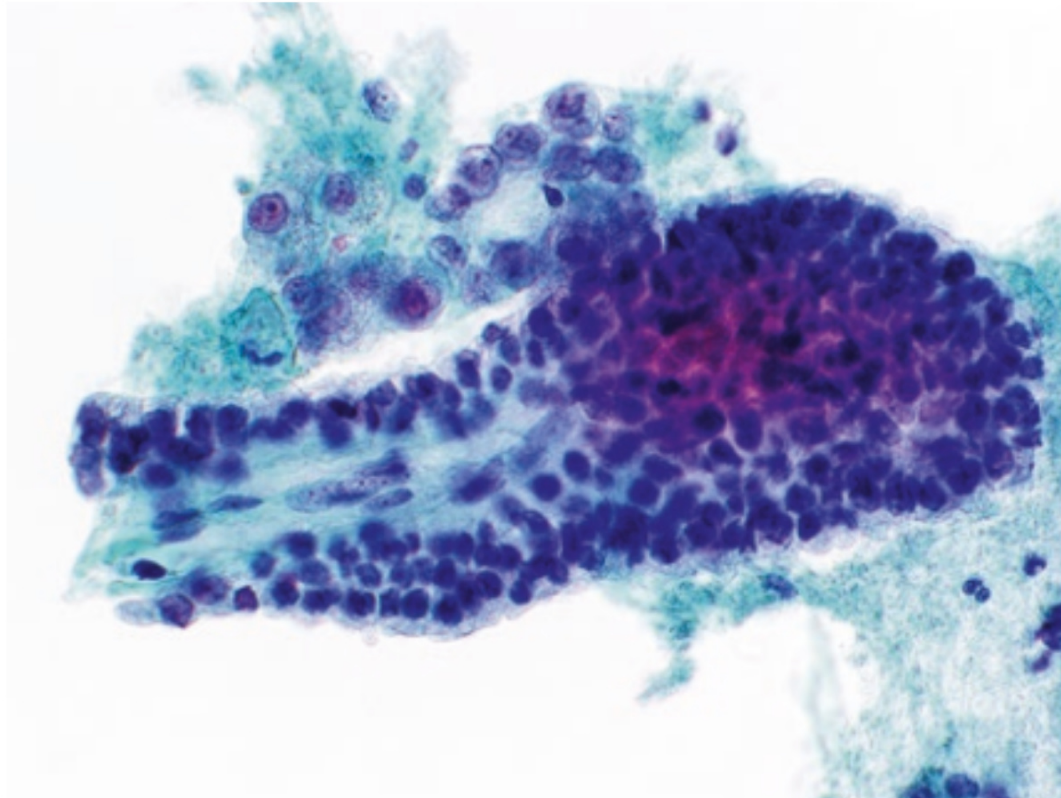
Fig. 8.8 Comparison of benign follicular cells with the cells of papillary thyroid carcinoma. (a) Benign follicular cells (nodular goiter). (b) Compared with those of the benign follicular cells, the nuclei of papillary carcinoma are larger, paler, more crowded, and more irregular in contour. (a and b, ThinPrep, Papanicolaou stain)

Papillary Thyroid Carcinoma

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Medullary Thyroid Carcinoma

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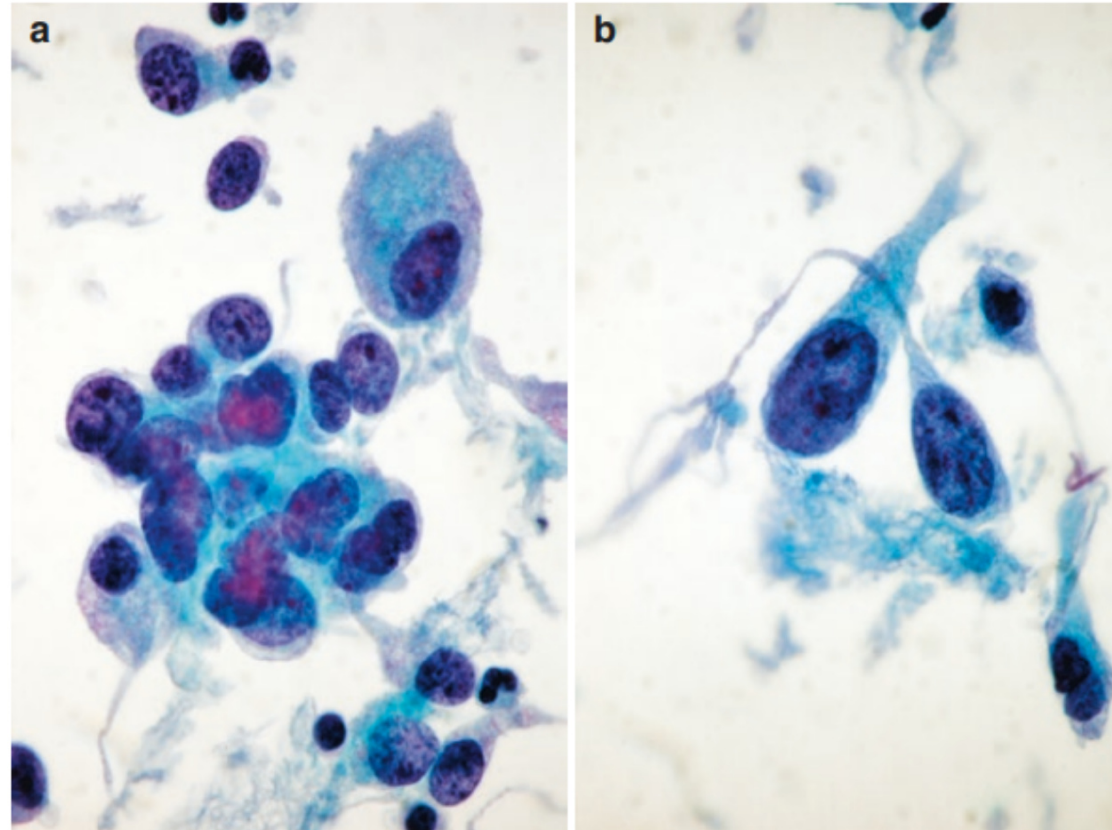


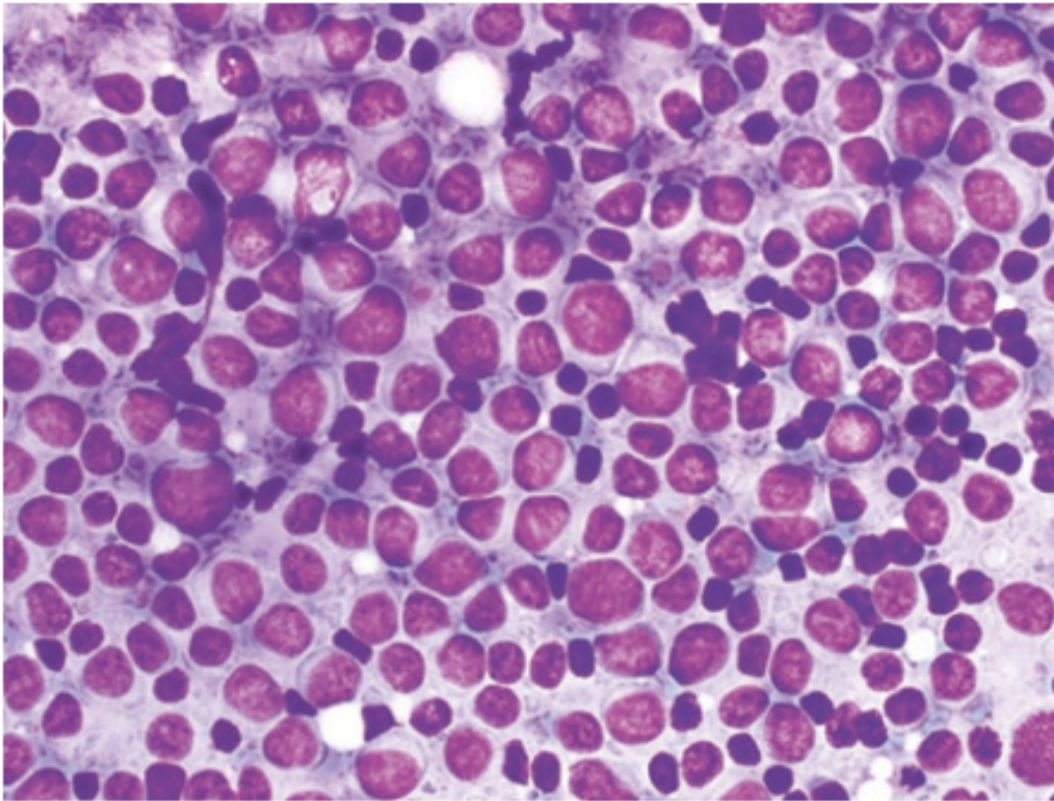
Fig. 9.3 Medullary thyroid carcinoma. A variety of shapes (round, polygonal, plasmacytoid (a), or spindled (b)) are seen in this population of tumor cells (smear, Papanicolaou stain)

Other Tumors

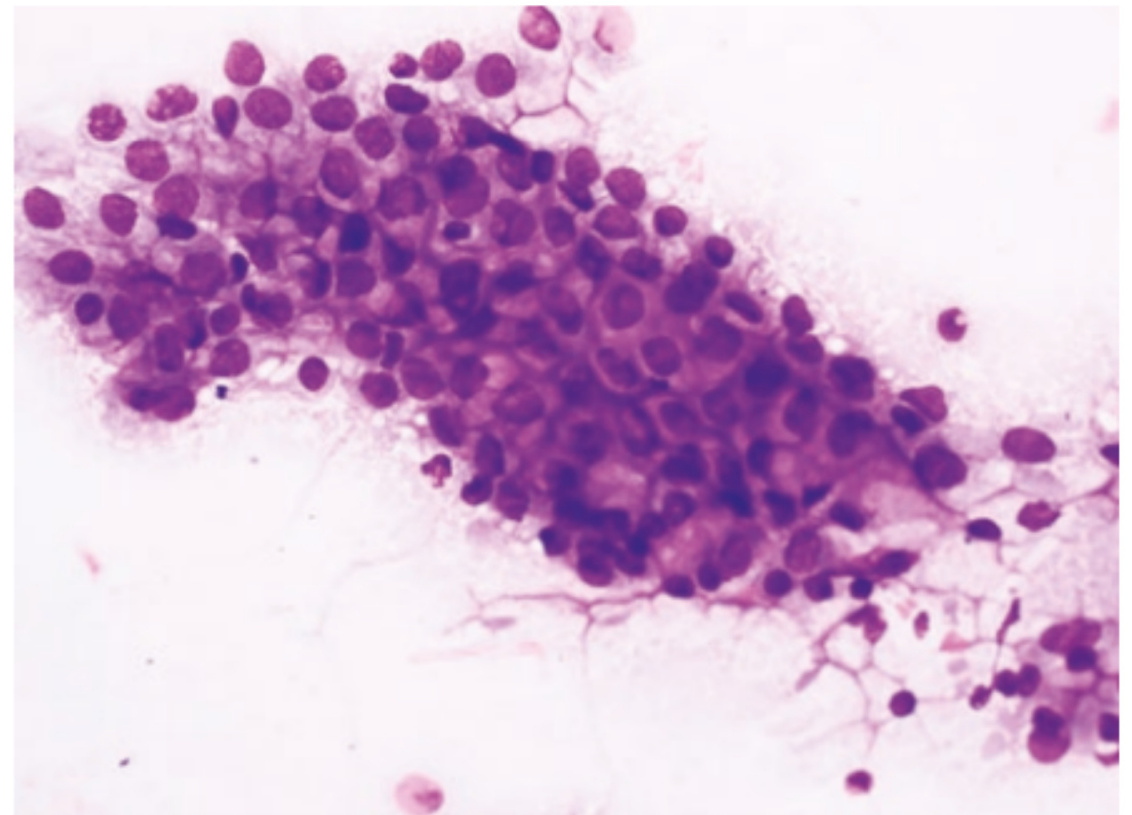
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Lymphoma



Metastatic Adenocarcinoma

Notes

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- FNAB cannot diagnose follicular carcinoma because in follicular carcinoma it requires demonstration of capsular invasion or vascular invasion which cannot be seen in FNAB.
- NIFTP cannot be **definitively** diagnosed by FNA; its diagnosis requires histologic examination of the resected lesion.
- Bethesda category SHOULD NOT become the sole determinant of management.
- Clinical relevances + ultrasound + Bethesda category = management

Terima Kasih

Source :

- Ali SZ, Baloch ZW, Cochand-Priollet B, et al. The 2023 Bethesda System for Reporting Thyroid Cytopathology. Thyroid. 2023.

